ITA0608 – Machine Learning Lab Practicals

EXP 07:

Write a program to implement Logistic Regression (LR) algorithm in python

Aim:

To implement the Logistic Regression (LR) algorithm in Python and classify data using an appropriate dataset.

Algorithm:

- Load the dataset and split it into training and testing datasets.
- Perform feature scaling to normalize the input data.
- Train the Logistic Regression model using the training data.
- Predict the class labels for the testing data.
- Evaluate the model using the Confusion Matrix, Accuracy, and classify a new sample.

Program:

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

iris = load_iris()

X = iris.data

y = iris.target

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.3, random_state=42

)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)

model = LogisticRegression(random_state=42)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
```

```python
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n")
print(cm)
print("\nAccuracy:", round(accuracy * 100, 2), "%")
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred, target_names=iris.target_names))
new_sample = [[5.1, 3.5, 1.4, 0.2]]
new_sample_scaled = scaler.transform(new_sample)
prediction = model.predict(new_sample_scaled)
print("\nNew Sample:", new_sample[0])
print("Predicted Class:", iris.target_names[prediction][0])
```

Output:

```
... Confusion Matrix:

    [[19  0  0]
     [ 0 13  0]
     [ 0  0 13]]

    Accuracy: 100.0 %

    Classification Report:

                  precision    recall  f1-score   support

          setosa       1.00      1.00      1.00        19
      versicolor       1.00      1.00      1.00        13
       virginica       1.00      1.00      1.00        13

        accuracy                           1.00        45
       macro avg       1.00      1.00      1.00        45
    weighted avg       1.00      1.00      1.00        45


    New Sample: [5.1, 3.5, 1.4, 0.2]
    Predicted Class: setosa
```

Result:

The Logistic Regression (LR) algorithm was successfully implemented in Python using the Iris dataset.